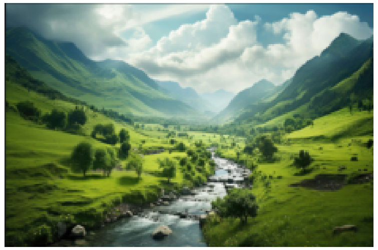
```python
# Upload Image
from google.colab import files
uploaded = files.upload()

import cv2
import numpy as np
import matplotlib.pyplot as plt

# Get the uploaded filename
filename = list(uploaded.keys())[0]

# Read the image
img = cv2.imread(filename)

# Check if image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Convert the image to grayscale
    gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Create a 5x5 rectangular kernel
    kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))

    # Perform erosion
    eroded_image = cv2.erode(gray_image, kernel, iterations=1)

    # Save the output image
    cv2.imwrite("Eroded_Image.jpg", eroded_image)

    # Display Original Image (Small Size)
    plt.figure(figsize=(3,3))
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis("off")
    plt.show()

    # Display Eroded Image (Small Size)
    plt.figure(figsize=(3,3))
    plt.imshow(eroded_image, cmap='gray')
    plt.title("Eroded Image")
    plt.axis("off")
    plt.show()
```

Choose files Screenshot … 105943.png

Screenshot 2026-07-22 105943.png(image/png) - 1074580 bytes, last modified: 22/07/2026 - 100% done

```
Saving Screenshot 2026-07-22 105943.png to Screenshot 2026-07-22 105943 (2).png
```

Original Image



Eroded Image